MILITARY SPECIFICATION

TRUCK, CRASH FIRE RESCUE, 3,000 GALLON CAPACITY, A/S32P-23

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers one type of diesel engine driven, crash fire rescue truck with a minimum capacity of 2,500 gallons of water.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

GGG-H-33	-	Hammer, Hand and Mallet, Surface Protective, Holders and Adapters, Inserted Hammer Face, and Faces, Hammer, Inserted.
PPP-C-96	-	Cans, Metal, 28 Gauge and Lighter
W-B-131	-	Battery, Storage (Vehicular, Ignition, Lighting and Starting).
L-S-300	-	Sheeting and Tape, Reflective, Nonexposed Lens.
GGG-W-665	-	Wrench, Spanner.
VV-F-800	-	Fuel Oil, Diesel.
GGG-A-926	-	Ax, (Single Bit, Double Bit, Pick Head, and Mattock Head).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: The Engineering Division, WR-ALC/MMEDO, ROBINS AFB, GA 31098-5094 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-P-514	-	Plates, Identification, Instructions and Marking, Blank.
MIL-T-5624	-	Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-P-23377	-	Primer Coating, Epoxy-Polyamide, Chemical and Solvent Resistant.
MIL-F-24385	-	Fire Extinguishing Agent, Aqueous Film- Forming Foam (AFFF), Liquid Concentrate, for Fresh and Sea Water.
MIL-B-38741	-	Bromocholordifluoromethane, Technical
MIL-C-46168	-	Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-C-52404	-	Connections: Hose, Fire and Water.
MIL-C-53039	-	Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant.
MIL-T-83133	-	Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.
MIL-C-83286	-	Coating, Urethane Aliphatic Isocyanate, for Aerospace Applications.

STANDARDS

FEDERAL

FED-STD-595	-	Colors.

MILITATY

MIL-STD-12	-	Abberviations for Use on Drawings, and on Specifications, Standards and Technical Documents.
MIL-STD-130	-	Identification Marking of US Military Property.
MIL-STD-461	-	Electromagnetic Emission and Susceptability Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurements of.
MIL-STD-721	-	Definitions of Terms for Reliability and Maintainability.
MIL-STD-781	-	Reliability Design Qualification and Production Acceptance Test: Exponential Distribuition.
MIL-STD-808	-	Finishes, Materials and Processes for Corrosion Prevention and Control in Support Equipment.
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines.
MIL-STD-831	-	Test Report, Preparation of.
MIL-STD-889	-	Dissimilar Metals.
MIL-STD-970	-	Standards and Specifications, Order of Precedence, for the Selection of
MIL-STD-1223	-	Nontactical Wheeled Vehicle Treatment, Painting, Identification Marking, and Data Plate Standards.
MIL-STD-1472	-	Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1595	-	Qualification of Aircraft, Missile and Aerospace Fusion Welders.

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MIL-STD-1791	-	Design for Internal Aerial Delivery in Fixed Wing Aircraft.
MS52131	-	Connector, Plug, Electrical Intervehicle Power Cable.

COMMERCIAL ITEM DESCRIPTION

A-A-393	-	Extinguisher, Fire, Dry Chemical (Hand Portable)
A-A-1108	-	Extinguisher, Fire, Vaporizing Liquid

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and other publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

LAWS AND REGULATIONS.

30CFR11	-	Respiratory Protection Devices; Test for Permissibility; Fees
49CFR393	-	Federal Motor Carrier Safety Regulations
49CFR571	-	Federal Motor Vehicle Safety Standards

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendant of Documents, US Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for the issuance thereof.)

DRAWINGS.

ARMY

11674728	-	Connector Assembly, Battery Booster
11682336-1	-	Cable Assembly

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by the contractor in connection with specific aquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME Boiler and Vessel Code, Section VIII, Pressure Vessels

(Application for copies should be addressed to the American Society of Manchanical Engineers, United Engineering Center, 345 East 47th Street,

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EUROPEAN TYRE AND RIM TECHNICAL ORGANISATION

Standards

(Application for copies should be assressed to the European Tyre and Rim Organisation, 32 Avenue Brugmann-B, 1060, Brussels Belguim.) New York, NY 10017.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 412	-	Evaluating Foam Fire Fighting Equipment on Aircraft Rescue and Fire Fighting Vehicles
NFPA 414	-	Aircraft Rescue and Fire Fighting Vehicles
NFPA 1931	-	Fire Department Ground Ladders
NFPA 1961	-	Standard for Fire Hose

(Application for copies should be addressed to National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J293	-	Vehicle Grade Parking Performance Requirements
J360	-	Motor Vehicle Grade Parking Performance Test Code
J366b	-	Exterior Sound Level for Heavy Trucks and Buses
J534	-	Lubrication Fittings
J695	-	Turning Ability and Off Road Tracking-Motor Vehicles
J786a	-	Brake System Road Test Code-Truck, Bus and Combination of Vehicles
J833	-	USA Human Physical Dimensions
J925	-	Minimum Service Access Dimensions for Off-Road Machines
J992b	-	Brake System Performance Requirements-Truck, Bus and Combination of Vehicles
J994	-	Alarm-Back Up-Electric-Performance, Test and Application
J1349	-	Engine Power Test Code-Spark Ignition and Diesel

(Applications for copies should be addressed to Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION

Yearbook

(Applications for copies should be addressed to Tire and Rim Association Incorporated, 3200 West Market Street, Akron, OH 44313.)

UNDERWRITERS LABORATORIES, INCORPORATED (UL)

ANSI/UL92	-	Standard for Fire Extinguisher and Booster Hose
ANSI/UL711	-	Classification Rating and Fire Testing

(Applications for copies should be addressed to Underwriters Laboratories, Incorporated, Publications Stock, 333 PFingsten Road, Northbrook, IL 60062.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersed applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Design.

3.1.1 Selection of specifications and standards. Specifications and standards, necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.1.2 Materials. Materials shall be as specified herein. Materials not specifically covered by this specification or referenced document shall be suitable in every respect for this type equipment. Except as otherwise specified, wood shall not be used.

3.1.2.1 Protective treatment. Materials that are subject to deterioation shall be protected in such a manner that shall in no way prevent compliance with the performance requirements of this specification. Protective coatings that will chip, crack, or scale with age or extremes of climatic conditions shall not be used.

3.1.2.2 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.1.3 Design and construction. The vehicle shall be constructed so that parts do not work loose in service and shall withstand the loads, shocks, vibrations and other conditions incident to operation, shipping and storage with minimum loss of time for maintenance, repair and servicing. The design of the vehicle shall ensure safe, efficient and economical operation. All assemblies, controls and installed equipment shall be located so that there is no adverse interference with each other, or with the operation, removal and replacement. The vehicle shall be capable of operating on paved surfaces and cross country terrain, and shall comply with all applicable requirements of the Federal Motor Vehicle Safety Standards, The Federal Motor Carrier Safety Regulations and NFPA 414, except as otherwise specified herein.

3.1.3.1 Reliability. The vehicle, including all components, shall have a Mean-Time-Between-Failures (MTBF) of not less than 100 hours demonstrated in accordance with Test Plan XVIIC of MIL-STD-781. A failure is defined as the event, or inoperable state, in which any item, or part of an item, does not, or would not, perform as specified. Definitions of reliability terms not included herein shall be as specified in MIL-STD-721 and MIL-STD-781. In the event of a conflict between these documents, MIL-STD-781 shall take precedence. The vehicle shall be designed for a life expectancy of at least 12 years.

3.1.3.2 Maintainability. The vehicle shall be designed and constructed (see 4.6.8) to provide the following:

a. A minimum number of parts consistent with the performance required herein.

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b. A minimum amount of time for assembling, disassembling, locating trouble sources and maintaining the vehicle. Where practical, components shall be located for rapid inspection and recognition of potential failure.

c. Ease of adjusting, servicing and replacing parts and components.

d. Use of readily available standard tools and equipment for maintenance.

e. Maintenance with a minimum number of tools.

f. Mean corrective maintenance downtime of no greater than one hour.

g. Maximum corrective maintenance downtime of no greater than four hours.

h. Maximum total downtime to perform all daily scheduled maintenance and servicing tasks of no greater than 35 minutes.

The vehicle shall be designed so that one mechanic, using common hand tools, can perform the following maintenance operations within the time specified:

(1) Remove, replace and adjust all engine driven belts (45 minutes).

(2) Remove and replace the alternator (30 minutes).

(3) Remove and replace engine coolant system hoses, except heater hoses (30 minutes).

(4) Drain engine lubricating oil, remove and replace oil filter element(s) and refill crankcase (1 hour).

(5) Remove and replace fuel filter element(s) (30 minutes).

(6) Disconnect battery cables, remove and replace batteries and reconnect battery cables (30 minutes).

(7) Drain torque converter oil and transmission oil, remove and replace all filter elements and strainers and refill converter and transmission (2 hours).

(8) Remove and replace starter (1 hour).

(9) Remove or install the engine/transmission or engine/power divider package, two mechanics (4 hours).

3.1.3.3 Foolproofness. Where improper installation of an item could cause a malfunction, an asymmetric mounting system shall be provided, whrer practical, to ensure proper mounting of the item.

3.1.3.4 Fastening devices. All screws, bolts, nuts, pins and other fastening devices shall be properly designed, manufactured and installed with adequate means of preventing loss of torque or adjustment. Cotter pins, lock washers or nylon patches shall not be used for this purpose, except for the attachment of trim items. Tapped aluminum threads shall have a minimum threaded engagement of at least two times the nominal fastener diameter.

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3.1.3.5 Servicing provisions. Drains, grease fittings, lines and check points for all components shall be so located that they are readily accessible and do not require special tools for proper servicing, removal, replacement and repair.

3.1.3.6 Lubrication. All parts normally requiring grease for lubrication shall be provided with lubrication fittings in accordace with SAE J534. Where the use of high lubrication pressures may damage grease seals, a pressure relief device shall be provided. Extended fittings shall be provided to lubricate parts or assemblies which are not readily accessible for direct lubrication, or which are likely to be overlooked because of inaccessibility.

3.1.3.7 Component protection. All space in which work is performed during operation, service and maintenance of the vehicle shall be free of hazardous protrusions, sharp edges or other features which may cause injury to personnel. All rotating and reciprocating parts and all parts subject to high operational temperatures or subject to being electrically energized, that are of such nature or so located as to be hazardous to personnel, shall be guarded or insulated to the extent necessary to eliminate the hazard.

3.1.3.8 Foreign object damage. All loose metal parts, such as pins and valve caps, shall be securely attached so they cannot become separated from the vehicle.

3.1.3.9 Bonding. Metal parts shall be specifically bonded together to minimize the possibility of sparks.

3.1.4 Environmental conditions. The vehicle shall be capable of staisfactory storage and operation under the following environmental conditions:

a. Full-time exposure to temperatures ranging from -40 degrees F to 125 degrees F.

b. Full-time exposure to relative humidities up to 100 percent.

c. Part-time (five percent of the time) exposure to rain; droplets of size 2.0 to 4.5 mm, falling at a rate of 5.2 inches per hour.

d. Part-time (five percent of the time) exposure to salt fog; five percent solution (five parts by weight of salt in 95 parts by weight of water) with fallout of 3.0 ml/80 cm2/hour.

e. Part-time (five percent of the time) exposure to blowing dust; velocity 1750 ft/min, concentration of 0.5 g/ft3.

f. Part-time (five percent of the time) exposure to blowing sand; velocity 5700 ft/min, concentration of 0.0773 g/ft3

3.1.5 Weight distribution. The Weight distribution of the vehicle in any loading condition shall not be unbalanced or improperly distributed. The difference in weight between the left and right wheels of any axle shall not exceed 2.5 percent of the total load on the axle. The difference in weight between axles shall not exceed 10 percent of the weight at the most heavily loaded axle.

3.1.6 Body lean. The vehicle shall not lean to either side more than one percent of the body width, in either the fully loaded or the unloaded condition. Body lean

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measurements shall be made at the outer most portion of the vehicle (excluding mirrors, mirror support brackets, lights or assymetrical shapes) directly above each axle; the difference in body height at each axle shall not exceed one percent of the body width at that location.

3.1.7 Electromagnetic interference. The vehicle shall comply with the requirements of MIL-STD-461, Class C1, Group I for electromagnetic interference, except that the limits for UM03 (Figure 8-1) shall be relaxed by 20dB.

3.1.8 Sound levels.

3.1.8.1 Exterior sound level. The maximum sound level produced by the vehicle with the agent system pump disengaged shall not exceed 86dB when measured in accordance with SAE J366b. An additional 2bBA allowance shall be permitted for production units to provide for variation in test site, temperature gradients, test equipment and inherent differences in nominally identical vehicles.

3.1.8.2 Cab interior sound level. The cab interior sound level shall be in accordance with Federal Motor Carrier Safety Regulation (FMCSR) 393.94.

3.1.9 Air transportability. The vehicle shall be air transportable in accordance with MIL-STD-1791 in a Type C-5 Aircraft and shall also be air transportable in a Type C-17 Aircraft. Preliminary design guidelines for the Type C-17 are 142 inch maximum height, 204 inch maximum width, 812 inch maximum length, 9 degree ramp angle and a floor load capacity equal to that allowed for the Type C-5 Aircraft. The air transportable configuration shall include all auxiliary equipment, foam, Halon 1211 and nitrogen tanks full, and water in the agent system, except that the water tanks shall be empty. Removal of equipment shall be permitted, if necessary, to provide the required height and width for air transport. Removal and reinstallation of equipment shall be limited to removal and attachment of standard fasteners and fittings with common hand tools. The weight of individual parts or assemblies which must be handled or positioned shall not exceed the limits specified in MIL-STD-1472 for a one person lift. All equipment removed shall be stored on the vehicle; all necessary caps and plugs shall be provided to permit driving and storage of the vehicle while in the air transport configuration. Removal or reinstallation of the equipment shall take no longer than 15 minutes for two mechanics using common hand tools.

3.1.9.1 Tiedowns. The vehicle shall be equipped with sufficient tiedown points so located that it can be restrained in all directions during air transport. The attachment points shall be designed for use with attachments in the aircraft as described in MIL-STD-1791. Sufficient clearance shall be maintained around all attachment points to allow for restraining chains without interference from any other structure. The tiedown points shall have a minimum clear opening of three inches to allow attachment of a chain with a grabhook.

3.2 Performance.

3.2.1 Mobility. The fully loaded and equipped vehicle shall:

a. Maintain a speed of 65 miles per hour (MPH) on a level paved road for continuous periods of at least one hour.

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b. Accelerate from a stop to a speed of 50 MPH within 40 seconds on a level paved road. Acceleration may be demonstrated with the engine fan disengaged.

c. Ascend and descend a 30 percent grade without spillage of any fluid, and a 50 percent grade with spillage of water and foam concentrate allowed.

d. Negotiate a 20 percent side slope in each direction while discharging agent at the maximum rated turret capacities.

e. Negotiate, in each direction, a ramp with a 36 percent minimum slope connecting two horizontal surfaces. The ramp length shall be at least 1.2 times the wheelbase of the vehicle. The vehicle shall be capable of being driven forward or backward from one horizontal surface across the ramp crest, ascend or descend the ramp, cross the ramp crest and proceed on the other horizontal surface without any portion of the vehicle, other than its tires, contacting any of the three surfaces.

f. Traverse graded gravel roads at a speed of 30 MPH.

g. Traverse Belgian Block at a speed of 20 MPH.

h. Traverse cross country terrain at a speed of 20 MPH.

i. Operate at speeds from one to ten MPH while discharging agent at maximum rated turret capacities. Pump engagement shall be possible with the vehicle moving at maximum speed, without interruption of tractive effort for a period longer than normally involved in the shifting of the transmission. Such engagement shall not result in a harsh reduction in vehicle speed or damage any component. With the agent pump engaged, it shall be possible to maneuver the vehicle without any interruption of agent system operation or reduction of agent system performance. While agent pump is engaged, control of vehicle speed shall not be by the brake system alone.

j. Ascend and descend a 40 percent grade while discharging at maximum turret capacities.

k. Comply with all service brake performance requirements of SAE J992b and FMVSS 121.

l. Comply with all parking brake performance requirements of SAE J293, except that the required grade shall be 20 percent.

m. Negotiate 14 inch high blocks with diagonally opposite wheels of the vehicle.

n. Have a wall to wall turning diameter of less than three times the vehicle length.

o. Traverse, in one pass at low speed, terrain consisting of unflooded fine grained dry to moist soils having a minimum cone index rating in the critical layer of no greater than 30. The vehicle cone index rating shall be measured in accordance with the Waterways Experimental Station Mobibity Index Method.

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p. Attain no more than six watts average vertical absorbed power at the driver's seat while negotiating a 0.7 inch RMS course at a speed of 25 MPH, a 1.0 inch RMS course at a speed of 17 MPH, and a 1.5 inch RMS course at a speed of 12 MPH with the tires at normal cross country inflation pressure. The vehicle shall exhibit no more than 2.5 G acceleration at the driver's seat while negotiating an eight inch high, half round obstacle at a speed of 12 MPH, and a ten inch high, half round obstacle at a speed of at least 7 MPH with the tires at normal cross country inflation pressure. Whole body vibration shall not exceed the values specified in MIL-STD-1472, Figure 42, for the time and frequencies specified.

The vehicle shall be capable of negotiating the tractive effort portions of the mobility requirements as described in 3.3.1(h), (o), and (p) using tires that have been worn to 30 percent or less of original tread depth.

3.2.2 Agent system performance.

3.2.2.1 Agent discharge. The agent system shall be capable of discharging water and foam selectively from the roof turret at flow rates of at least 750 and 1,500 gallons per minute (GPM), from the bumper turret at a flow rate of at least 250 GPM and from a handline at a flow rate of at least 60 GPM, using foam concentrate in accordance with MIL-F-24385. The foam shall have an expansion rate of at least three and be capable of obtaining a 25 percent drainage time for each turret of two minutes and one minute for the handline. All specified flow rates are for unexpanded foam. The vehicle shall allow selective operation of the roof turret, the bumper turret and the handline in any combination with assurance that agent, proportioned within specified limits, is available at the inlet of each nozzle within five seconds after foam has been selected, regardless of the level of water and foam concentrate in the agent tanks. All discharge patterns shall be homogeneous, symmetrical and continuous. Each nozzle shall distribute agent at a uniform application rate (within a tolerance of plus or minus 10 percent) over at least 80 percent of the total pattern area when discharging any selected pattern with the nozzle elevated for maximum range. The specified performance shall be achieved without assistance from the wind. Discharge distance shall be measured from the tip of the nozzle to the far side of the pattern. Pattern distance and width shall be measured at maximum range. All dimensions are in feet.

ROOF TURRET PERFORMANCE

	WATER DISCHARGE		FOAM DISCHARGE	
	Distance (min)	Width (min)	Distance (min)	Width (min)
Straight Stream	250	N/A	250	N/A
Dispersed	75	35	75	35

3.2.2.1.1 Agent discharge connection performance. The agent system shall be capable of discharging water and foam selectively from each 2-1/2 inch discharge connection at a flow rate of at least 250 GPM. Total agent discharge flow rate for all outlets (including turrets) shall be 1,810 GPM minimum; all outlets are not required to discharge simultaneously.

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3.2.2.2 Dry pump operation. The agent pump shall be capable of dry operation at maximum rated speed for 10 consecutive minutes without sustaining damage or reduction in performance.

3.2.3 Halon 1211 system performance. The Halon 1211 system shall be capable of extinguishing three 600 square foot liquid fuel fires in accordance with UL 711, without recharge of agent or nitrogen.

3.3 Chassis.

3.3.1 Powertrain.

3.3.1.1 Engine. The vehicle shall be equipped with a commerical diesel engine of not less than six cylinders. The total power output required by all items driven by the engine shall not exceed its net power output rating at 2,000 feet elevation, determined in accordance with SAE J1349. The engine shall demonstrate the performance characterisitics specified herein when using diesel fuel in accordance with VV-F-800, and shall operate satisfactorily on Grades JP-4 and JP-5 turbine fuel in accordance with MIL-T-83133. Any engine that is adversely affected by any of the above fuels is not acceptable.

3.3.1.1.1 Engine installation. The engine shall be located in a compartment so as to provide maximum accessibility for service and maintenance from both sides, in front of, above and below the engine without removing components other than underbody cowling.

3.3.1.1.2 Engine lubrication system. The engine lubrication system shall function satisfactorily on grades up to 50 percent and side slopes of up to 30 percent. The oil filiter shall be the engine manufacturer's standard spin on type. An oil filler, designed to allow oil to be readily poured into the filler opening from a five quart container conforming to Type III of PPP-C-96 without the use of a funnel, shall be provided.

3.3.1.1.3 Engine cooling system. The engine shall be equipped with a cooling system capable of satisfactory performance under the environmental conditions specified herein; coolant temperature shall not exceed the engine manufacturer's recommended maximum. Expansion tanks of sufficient capacity for expansion of all coolant in the system shall be provided. The engine shall be equipped with a thermostatically controlled fan.

3.3.1.1.4 Engine starting system. The engine shall start within 15 seconds in any ambient temperature within the required operating range of the vehicle. Installed glow plugs, fluid starting aids and heat from the winterization system (see 3.6) may be used prior to and during the start period to facilitate engine starting under the following conditions:

TEMPERATURE RANGE (DEGREES F)	STARTING AIDS PERMITTED
40 through 125	None
1 through 39	Glow plugs and fluid starting aids.
-40 through 0	Glow plugs, fluid starting aids and heat from the winterization system.

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The starting system shall be equipped with an interlock device to prevent starter engagement when the engine is running or when the transmission is in any position other than neutral.

3.3.1.1.5 Exhaust system. The exhaust system shall be designed to expel exhaust gases in a manner that does not contaminate any area within or surrounding the vehicle, create a safety hazard to personnel, or damage any component of the vehicle finish. The outlet shall be directed toward the ground. The exhaust system, including all fasteners, shall be constructed of corrosion resistant materials. The system shall be protected from damage which could result from traversing rough terrain. Provisions shall be incorporated to prevent water or agent from entering the system. The exhaust system shall be in accordance with FMCSR 393.83.

3.3.1.2 Transmission. A commerical automatic transmission shall be provided, complete with a hydraulc torque converter (see 6.3.2). The normal drive range selector position shall provide not less than four gear ratios without selector movement. The transmission shall have a torque capacity of at least 100 percent of the maximum net engine torque. The transmission and its controls shall be in accordance with FMVSS 102.

3.3.1.3 Interaxle differentials. Interaxle differentials shall be provided to compensate for differences in speeds between each axle. An automatic or operator controlled locking device shall be incorporated into each interaxle differential. The operator control, if provided, shall be located in the cab.

3.3.1.3.1 Axles. The load rating of each axle shall not be less than the imposed load on the axle, measured at the ground, with the vehicle at gross weight. Axle torque capacity shall be sufficient to spin the tires under all possible service conditions, including a 1.0 coefficient of friction. Axle design shall be such that, in the event of shaft failure, the wheel is not released from the vehicle. Maximum interchangeability of axle components shall be provided. Each axle shall be equipped with an automatic or operator controlled differential locking device. The operator control, if provided, shall be located in the cab.

3.3.2 Steering system. The vehicle shall be equipped with a power assisted steering system, with manual steering available in the event of power assist failure. The steering system shall not require more than five turns lock to lock. The force required for turning the wheels from lock to lock, with the vehicle at gross load and the engine at low idle, shall not exceed 15 pounds when measured at the steering wheel rim.

3.3.2.1 Steering wheel. The steering wheel shall have a diameter of 18 inches minimum. Not less than four inches clearance shall be provided between the steering wheel rim and the nearest projection in the plane of the steering wheel. Not less than 2 1/2 inches clearance shall be provided between the steering wheel rim and the nearest projection below the steering wheel.

3.3.3 Brakes.

3.3.3.1 Service brakes. The vehicle shall be equipped with full air brakes on all wheels. Brakes shall be disc or drum type. A commerical air dryer and necessary associated parts shall be added to the pneumatic system to eliminate excessive moisture and oil, thus increasing the life of the air brake components. It shall use desiccant for the final absorption process and shall be installed in

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the discharge line between the compressor and the first reservoir. A heater and thermostat assembly shall be provided with the air dryer to prevent freeze-up when used in severe winter conditions. A drain shall be provided on each air reservoir and shall be readily accessible for service. A pull away air connection for charging of air tanks from an external air source shall be provided, mounted at the rear of the vehicle. The service brakes shall be in accordance with FMVSS 121.

3.3.3.2 Parking brakes. The vehicle shall be equipped with parking brakes of the spring applied type on all nonsteering wheels; the parking brakes shall be in accordance with FMVSS 121.

3.3.4 Frame. The vehicle shall have a frame assembly adequate to withstand the loads imposed during the operations described herein. The main frame rails shall be one piece channel or rectangular tubing and shall extend from the most forward suspension mounting bracket to the rear bumper. The frame shall be assembled with SAE Grade 8 fasteners; no rivets or welding shall be permitted. The resisting bending moment (RBM) of each frame rail shall be at least [(0.175) (wheelbase) (gross vehicle weight)].

3.3.5 Suspension system. The vehicle shall be equipped with a suspension system adequate to provide complete control to the driver, prevent damage to components and furnish acceptable ride qualities under all specified operating conditions. When provided, springs shall be designed for at least 110 percent of the imposed weight.

3.3.6 Wheels and tires. The vehicle shall be equipped with wheels and tires which meet all Tire and Rim Association or European Tyre and Rim Technical Organisation requirements for this application. A size and type tire which is not currently listed by either of these organizations may be proposed, provided that a copy of the application for inclusion and engineering data sufficient for evaluation are provided. The size and type tires shall be listed by one of these organizations prior to first article testing. Tubeless steel belted radial ply tires with all terrain type tread shall be provided. Wheels shall be disc type. Front and rear tires shall be identical and interchangeable; disassembly of the tire and wheel assembly is allowed. Each wheel shall be provided with a molded rubber reinforced sleeve (bead lock). A spare wheel and tire assembly shall be provided with, but not mounted on the vehicle. Tires shall be in accordance with FMVSS 119; except for tread wear indicators and the tire markings. Tire and rim selection shall be in accordance with FMVSS 120.

3.3.6.1 Central tire inflation system. The vehicle shall be equipped with a driver controlled central tire inflation system (CTIS) having a minimum of three preset pressures: highway, cross country and mud/snow. The system shall be capable of reducing the infaltion pressure of all tires from highway pressure to mud/snow pressure within 30 seconds with the vehicle in motion. Reinflation from mud/snow pressure to highway pressure shall not require more than 15 minutes with the engine at maximum rpm. With the system in operation, tire pressure shall be verified and adjusted at intervals necessary to assure that no more than a 3 psi variation exists between selected pressure and actual pressure except during the inflation/deflation operation caused by the selection of a new tire pressure. The CTIS shall incorporate sufficient safeguards to assure that air pressure necessary to continue safe operation of the vehicle shall be available at all times during activiation of CTIS or in the event of CTIS failure. The system shall provide for the isolation of any or all tires from the CTIS in the event of CTIS failure for any reason. Valves for manual inflation shall be readily accessible.

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3.3.6.1.1 Sensors. The CTIS shall include sensing of the vehicle speed and comparing indicated speed to the maximum allowable speed for each control setting. In the event that vehicle average speed exceeds maximum allowable speed for that setting for a period of more than one minute, a panel mounted flashing amber light shall activate to warn the driver of this condition, and the system shall automaticall inflate to the appropriate pressure. Sensors shall alert the operator by a flashing amber light and activate the automatic inflation system whenever speed exceeds that suitable for sustained operation at pressures appropriate to mobility requirements.

3.3.7 Fuel system. The fuel system shall be in accordance with FMCSR 393.65.

3.3.7.1 Fuel tanks. Fuel tanks shall be provided with a capacity of at least 100 gallons. Each tank shall have a fill opening of three inches minimum readily accessible to personnel standing on the ground and designed to prevent fuel splash while refueling. The tank(s) shall be located and mounted so as to provide maximum protection from damage. If more than one tank is furnished, means shall be provided to assure equalized fuel level in both tanks.

3.3.7.2 Fuel filter. At least one filter of the fuel-water separator type shall be located in the fuel line between the fuel tank(s) and the engine. It shall be installed so that the element can be readily serviced without removing surrounding equipment. Unless the filter is mounted higher than the fuel tank(s), a shutoff valve shall be included in the inlet line to the filter.

3.4 Cab. The vehicle shall be equipped with a fully enclosed cab with seating for a driver, a turret operator and two additional crew members. The driver and the turret operator shall be seated beside each other near the front center of the cab. The additional crew seats shall be located outboard of the driver and turret operator seats; the crew seat backs shall be at least 10 inches aft of the driver and turret operator seat backs. Body dimensions shall be as specified in SAE J833 for all personnel falling within the range of a small man (clothed) through a large man (artic clothed), allowing two additional inches for height due to fire protective clothing. Not less than 12 inches headroom shall be provided. Unobstructed passage shall be provided to each seat. Grab bars shall be furnished as needed to provide entry into the cab and stability for crew members during cross country operation. Components projecting inside the cab shall be padded. A space at least 18 inches long by 12 inches wide by 12 inches high shall be provided for mounting a radio for base operations (see 3.6.4). A location for mounting the radio controls shall be provided adjacent to the siren controls (see 3.6.5), such that they are readily accessible to the driver and the turret operator. Space for bunker clothing for four persons shall be provided behind the forward seats in a location accessible to the crew during travel. The cab shall comply with the visibility requirements of NFPA 414 and shall be equipped with with at least two sun visors. A quick opening passage providing access to the roof turret in not required unless the manual override is located at the roof turret. Any piping in the cab shall be completely covered by panels which are readily removable for service. The colors of the interior finish shall be approved by the procuring activity.

3.4.1 Seats. Confortable nonsuspension bucket type seats shall be provided at all seating positions. Seats for the driver and the turret operator shall have a minimum horizontal and vertical adjustment of four inches. The seat backs of each outboard seat shall incorporate a recessed quick-release holder for a strap on a self-contained breathing air apparatus, designed so that the crew member can attach

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the apparatus while seated. Upholstry shall be heavy duty black vinyl. Inertia reel seat belts shall be provided, attached to extensions that position the belt ends beside the seat. Seats shall be in accordance with FMVSS 207; seat belts and their installation shall be in accordance with FMVSS 209 and FMVSS 210.

3.4.2 Doors. The cab shall be equipped with at least one door on each side, sized and located for convenient entry and exit of operating personnel. Door handles shall be designed for operation by personnel wearing artic mittens as specified in SAE J925, with the latch designed to prevent entry of water and ice into the mechanism. If piano type hinges are provided, they shall be of stainless steel. The upper portion of each cab door shall be provided with an opening roll down type or sliding type window. A nonopening window shall be provided in the lower portion of each door to provide increased visibility. Door retention shall be in accordance with FMVSS 206.

3.4.3 Windshield and windows.

3.4.3.1 Windshield, wipers and washers. The windshield shall be of tinted safety glass and shall be the maximum width consistent with the vehicle. The vehicle shall be equipped with electrically powered windshield wipers designed to provide clear vision for the cab occupants during rain, snow and fire fighting operations. The wipers shall clear at least 75 percent of the windshield aera, have a minimum of two speeds and shall be stowed out of the operator's line of vision when not in use. A windshield washer with a minimum capacity of two gallons shall be provided; the reservoir shall be constructed of unbreakable material. In addition, a cooling spray from the water tank shall be provided to cool the windshield during fire fighting operations. The spray shall shall be operable reguardless of whether the agent pump is engaged, and shall not contain foam under any operational condition. The windshield shall be in accordance with FMVSS 205. The wipers and washers shall be in accordance with FMVSS 104.

3.4.3.2 Windows. Tinted safety glass windows of a size to provide maximum practical visibility shall be installed in the cab and doors. The windows shall be in accordance with FMVSS 205.

3.4.4 Mirrors. An adjustable rearview mirror of not less than 90 square inches shall [illegible] side of the vehicle on a retractable bracket; when specified (see 6.2), it shall be heated. The mirrors shall be positioned so that the driver can clearly observe the roadway and vehicles approaching from the rear on either side. A spotter mirror, at least five inches in diameter, shall be mounted below each rearview mirror. If a cab forward configuration is provided, a spotter mirror shall be mounted on each side of the windshield, installed so that the front of the vehicle is visible to the driver. All mirrors shall be constructed of glass and shall be mounted so as to resist vibration during vehicle operation; mirrors shall not be mounted on the doors. The mirrors shall be in accordance with FMVSS 111.

3.4.5 Instruments and controls. Instruments and controls shall be located and arranged so that those which are most vital to operation have the most predominate location. They shall be placed in blocks distinctly marked to group those instruments and controls pertinent to the operation of any component. For example, all instruments and controls relative to the operation of the engine shall be in one block. Controls having similar functions shall have similar actions. Indicator lights shall be green to indicate proper operation, amber to indicate caution and

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red when warning is intended. The headlight high beam indicator may be blue. All instruments and controls shall be illuminated with nonglare lighting to be readily visible for night operation. A circuit shall be provided for testing all indicator light bulbs. The instruments and controls shall be in accordance with FMVSS 101. The accelerator control system shall be in accordance with FMVSS 124. At least the following instruments and controls shall be provided:

a. Speedometer and odometer.

b. Tachometer with combination engine hourmeter.

c. Engine oil pressure gauge.

d. Engine coolant temperature gauge.

e. Transmission oil temperature gauge.

f. Voltmeter and ammeter.

g. Fuel level gauge.

h. Central tire inflation system controls.

i. Tire inflation pressure and status indicator lights.

j. Air pressure gauge with audible low pressure warning and warning light.

k. Water tank level sensor with four light indicator and low level warning alarm.

l. Foam tank level sensor with four light indicator and low level warning alarm.

m. Agent system engagement control.

n. Agent system engagement indicator.

o. Agent pump discharge pressure gauge.

p. Halon 1211 system charge indicator.

q. Headlight high beam indicator.

r. Siren controls.

s. Beacon control switch.

t. Master electrical switch.

u. Winterization system controls.

v. Central breathing air system pressure gauge with audible low pressure warning and warning light.

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3.4.6 Climate control system. The climate control system shall provide heater, defroster and air conditioner modes. The system shall include a heater core, a blower, a refrigerant compressor, a condenser, an evaporator and the plumbing, ductwork, dampers, controls and other components necessary to meet the performance requirements specified herein. The climate control system shall induct at least 60 cubic feet per minute of fresh air into the cab. At least four adjustable panel louvers shall be provided, approximately evenly spaced across the width of the cab; each shall provide approximately equal air flow. Cab mounted components shall be protected from inadvertent damage by personnel.

3.4.6.1 Heater and defroster. The heater shall maintain a temperature of at least 40 degrees F at the floor with an outside ambient temperature of -40 degrees F. When occupied by four crew members, the defroster shall keep at least 75 percent of the windshield and side windows clear with an outside ambient temperature of -40 degrees F. The defroster shall be in accordance with FMVSS 103.

3.4.6.2 Air conditioner. The air conditioner shall, at normal engine idle speed (standby condition), be capable of reducing the cab interior temperature to 95 degrees F within 30 minutes with the outside ambient temperature of 125 degrees F. The evaporator condensate shall drain to the ground under any operating condition.

3.4.7 Central breathing air system. A central self-contained breathing air system shall be provided complete with a full face mask with integral voice pack and a quick disconnect coupling at each seating position. The face mask shall provide adequate clearance for a crew member wearing eyeglasses. The system shall have a capacity of at least one hour of breathing air for each crew member (40 liters per minute per crew member). It shall include a breathing air cylinder, pressure demand regulator(s), high pressure hose and all required fittings and hardware. The system shall be installed so as to prevent damage during any operational condition describe herein. The air cylinders shall be firmly mounted during operation, but easily removable by two fire fighters for service. The breathing air system shall be approved by the National Institute for Occupational Safety and Healt (NIOSH) and the Mine Safety and Health Administration (MSHA), US Department of the Interior, under 30 CFR 11. It shall further be approved for use by the Air Force Occupational Environmental Health Labortory, Brooks AFB, TX.

3.4.8 Drain troughs. Drain troughs shall be provided on the vehicle to carry away all agent without spilling over or running into cab windows, over the windshield or compartment doors. The troughs shall not drain onto thoroughfares and locations anticipated to be occupied by personnel during normal operation.

3.5 Electrical system. The vehicle shall be equipped with a 12 volt electrical system; a 24 volt starting system may be provided. Wiring shall be weather proof and color coded or numbered for easy identification; wiring and its installation shall be in accordance with FMCSR 393.27, 393.28 and 393.33. All circuits shall be protected by circuit breakers located in the cab, a dedicated compartment or in the Halon 1211 hose reel compartment so as to be readily accessible to a mechanic; all circuit breakers shall be in accordance with FMCSR 393.31. A master switch shall be provided to disconnect the batteries from the electrical system.

3.5.1 Alternator. The engine shall be equipped with an alternator sufficient for operating all vehicle electrical components simultaneously, except for the hose reel motors (see 3.8.9, 3.9.3). The system shall supply full power requirements for operation of the truck and all equipment and accessories at engine idle speed

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without system degradation. An engine high idle speed control which automatically increases the engine speed may be provided. The device shall be preset so that when activated, it will operate the engine at the appropriate speed; maximum engine speed shall not exceed 1,200 RPM. The design shall allow for disengagement when the transmission is engaged or the parking brake released. Minimum capacity of the alternator shall be 160 amperes.

3.5.2 Batteries. The vehicle shall be equipped with at least two 12 volt batteries, rated at a total of 1,800 CCA minimum. The batteries shall comply with the maintenance free characteristics of W-B-131. Batteries shall be so arranged and connected to provide sufficient capacity for engine starting in all conditions specified herein. The batteries shall be mounted in an acid-resistant tray; a sliding carriage shall be provided if the batteries must be moved sideways to permit access for servicing or replacement. Battery installation shall be in accordance with FMCSR 393.30.

3.5.3 Lighting.

3.5.3.1 Vehicle lighting. The vehicle shall be equipped with headlights, tail lights, clearance lights, brake lights, turn signal lights and reflectors in accordance with FMVSS 108. Two red flashing warning lights, three inches in diameter minimum or equivalent rectangular size, shall also be provided. The warning lights shall be installed on the front of the vehicle in a protected location. The turn signal operating unit shall have visible and audible flash indicators; turn signals shall be self-cancelling; the headlamp dimmer switch shall be incorporated in the turn signal lever. Lights and reflectors shall not be mounted on the vertical surface of rub rails or on vehicle bumpers. One work area illumination light shall be installed on each side of the body. The light shall provide lighting at ground level to all work areas to each side of the vehicle. Each work area light shall be provided with a control switch located at the light area and a master control switch located on the cab instrument panel. A standby position shall be provided in which the parking and clearance lights operate. Nonglare white lighting shall be provided at all ladder step areas and any area where personnel may be required to work during operation. The ladder/step lighting shall be controlled by switches located on the instrument panel and at the ladder/step area. Lenses for all lights shall be of heat resistant material to prevent damage when near a fire.

3.5.3.2 Interior lighting. The cab and all compartments shall be provided with white lighting adequate to enable personnel to readily obtain equipment and perform all functions necessary during night operation. Sealed contact switches shall be provided to automatically illuminate the lights in the cab or a compartment when its door is open.

3.5.3.3 Spotlights. Spotlights shall be designed for a vertical angle movement of at least 60 degrees above and 35 degrees below the horizontal. All moving parts of the spotlight assemblies shall be lubricated and sealed.

3.5.3.3.1 Front spotlights. A spotlight shall be installed at the top of the vehicle at each of the front corners, designed to be aimed from inside the cab. A control switch shall be located on the control handle of each spotlight.

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3.5.3.3.2 Rear spotlights. A spotlight shall be installed at the top of the vehicle at each of the rear corners. The control handle for each rear spotlight shall be located on the vehicle below the light, not more than 48 inches above the ground. Spotlight switches shall be located in the control handle and on the cab instrument panel. Each spotlight shall be designed for 360 degree horizontal rotation.

3.5.3.4 Beacons. Two red strobe type beacons shall be installed, one fore and one aft. The location of the lights shall provide unobstructed view of the beacons to aircraft and personnel standing on the ground at a distance of not more than 25 feet from the vehicle. A signal switch on the instrument panel shall be provided to control the beacons.

3.5.3.5 Roof turret lights. A halogen spotlight shall be installed on each side of the roof turret of the vehicle, mounted to aim automatically with the movement of the turret nozzle. A switch located on the instrument panel shall be provided to operate the lights.

3.5.4 Communication equipment circuit. A dedicated electrical circuit with an individual circuit breaker shall be provided with wiring to the space provided for the radio installation (see 3.4). The circuit shall be rated at 30 amperes minimum.

3.5.5 Siren. An electronic siren shall be provided, complete with a speaker with a 100 watt minimum rating. The siren shall provide slow wail and simulated air horn modes. The siren shall include all necessary wiring so that it is adaptable to two way radios which use either carbon or magnetic type microphones, so that it can be used for a public address system and for broadcasting incoming radio messages through the siren speaker. Hand operated controls for actuating the siren shall be mounted on or adjacent to the instrument panel in a location (see 3.4) that is readily accessible to the driver and the turret operator.

3.6 Winterization system. The vehicle shall be equipped with a winterization system to permit storage and operation in an ambient temperature as low as -40 degrees F. Heaters and a built-in battery charger shall operate on 110 volt alternating current (AC) and shall be wired through a junction block to a single three pronged (male) weatherproof slave receptacle at the rear of the vehicle, designed for pull away connection, for receiving external power and grounding the vehicle. A three wire connecting cable, 25 feet long and of adequate line capacity to supply power for all heater units simultaneously, shall be furnished. The connecting cable shall include a matching female connector at the vehicle end and a standard weatherproof three pronged (two power plus one ground) male connector at the other end. Electrical appatatus shall conform to FMCSR 393.77 (c) (7). Electrical insulation of the connecting cable shall withstand normal operating stresses in low ambient air temperature without cracking or loss of dielectric capacity. All heater wires shall be installed without interfering with vehicle component operation and without loose excess wire. The following heaters, if required, shall be furnished:

a. Coolant heater, 1,500 watt minimum rating, shall be installed in the engine block or in the lower coolant inlet hose. Engine thermostat, with an opening range of 140 degrees to 160 degrees, shall be installed.

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b. Immersion type engine oil heater, 300 watt minimum rating, with 140 degrees to 160 degrees F thermostat, shall be installed in the oil pan through any convenient opening.

c. Battery heater shall be provided, having a capacity adequate to maintain the battery electrolyte at a temperature of not less than 10 degrees F during vehicle exposure in ambient temperatures as low as -40 degrees F, and shall embody a thermostat to limit the temperature of the electrolyte to not more than 80 degrees F.

d. A fuel warmer or preheater shall be provided to prevent clogging of fuel filters due to wax crystallization in the fuel. The fuel warmer shall use engine coolant to transfer sufficient heat to the diesel fuel to heat it from an ambient temperature of -40 degrees F to an outlet temperature of 9 degrees F, with a flow rate not less than the maximum fuel demand of the engine fuel system. A coolant shutoff valve shall be provided for the coolant inlet side of the fuel warmer unit.

The vehicle shall also be equipped with a 12 volt slave receptacle in accordance with Drawing 11674728, wired so as to allow starting of the engine. The receptacle shall be mounted adjacent to the AC receptacle. A power cable, in accordance with Drawing 11682336-1, shall also be provided. In cab storage shall be provided for the 110 volt and 12 volt cables.

3.6.1 Agent system winterization. The winterization system (see 3.6) shall include the necessary equipment required to prevent damage to any component of the agent system in low ambient temperatures. The agent winterization system shall be powered by the vehicle electrical system and shall be designed to provide the required performance regardless of engine speed. The system shall also provide the required heating when power is supplied from an external 110 volt AC source. The system design and installation shall include the necessary coolant flow shutoff features to permit removal and reinstallation of the major components of the system without requiring the coolant to be drained from the system and without coolant loss in excess of the capacity of the component being removed. The system shall be designed and the vehicle shall be insulated so that after stabilization of operating temperatures, the system, in an ambient temperature as low as -40 degrees F, can be shut down for a period of at least two hours without draining of the agent system and without damage to the vehicle. Incorporation of the winterization booster heaters shall not detract from the performance of the vehicle in normal or high ambient temperatures. The winterization system shall include, but not be limited to, the following components controlled by the driver:

a. Booster heater capable of operating on all fuels specified in 3.3.1.1.

b. Agent system heat exchangers.

The agent winterization system shall automatically shut down in the event of system failure; audible and visual warning shall be provided. All components shall be located for ease of removal, replacement and servicing.

3.7 Body. The vehicle shall be equipped with a body to enclose the foam and Halon 1211 agent systems and to provide storage for the auxiliary equipment listed (see 3.10). The framework shall be of welded aluminum construction with aluminum or fiberglass composite boby panels. The body shall have weathertight compartments with bottoms flush and level with, or slightly above, the bottoms of the doors. Doors fc

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11a. WORK TELEPHONE NUMBER (Include Area ... 11b. NAME OF SUBMITTER (Last, First, MI) - Optional

the foam hose reel compartment, the Halon 1211 hose reel compartment, and all storage compartments shall be of the roll up type; access doors may be hinged at the side or at the top. All doors shall be equipped with automatic devices to hold them in the open position and latches that are operable with one hand by a person wearing arctic mittens. Hinged doors shall swing through an arc of at least 150 degrees; full length piano type hinges with stainless steel pins shall be provided. Seals shall be provided to prevent water from entering through the hinges. Fasteners shall be recessed or flush with the compartment interior panels. At least one additional compartment adequate for storage of a box 36 inches long by 32 inches high by 24 inches deep shall be provided one on each side of the vehicle. Underbody panels shall be attached with quarter turn or other suitable quick release fasteners.

3.7.1 Catwlaks and deckplates. Catwalks and deckplates shall be furnished to provide access to all equipment mounted on the top of the vehicle. They shall be raised at least two inches above the surrounding surfaces and shall be constructed of non-skid aluminum grating or mesh. The catwalks and deckplates shall support a 250 pound crew member with no noticeable deflection. Catwalks shall be a minimum of 18 inches wide.

3.7.2 Steps and ladders. The height between steps or ladder rungs shall not excees 20 inches. Steps and ladder rungs shall be a minimum of 18 inches wide, except for the bottom step, which may be a minimum of eight inches wide. The bottom step shall be no higher than 22 inches from the ground when the vehicle is fully laden. Steps shall have a minimum depth of six inches. Steps and ladder rungs shall support a 250 pound crew member with no noticeable deflection.

3.7.3 Reflectors. The vehicle shall be equipped with reflectors in accordance with FMCSR 393.12 and 393.26.

3.8 Agent system.

3.8.1 Agent tanks. The vehicle shall be equipped with a water tank with a rated capacity of at least 3,000 gallons and a foam concentrate tank with a rated capacity of at least 450 gallons. Each tank shall be designed so that its rated capacity can be discharged at maximum rated flow through the roof and bumper turrets with the vehicle on level ground, and 75 percent of its rated capacity can be discharged with the vehicle facing either direction on a 20 percent side slope or on a 30 percent grade.

3.8.1.1 Agent tank constructions. The agent tanks shall be constructed of passivated 316L stainless steel or reinforced fiberglass; all materials used shall be certified as being compatible with water, foam concentrate and water/foam solutions. Each tank, with all covers in place, shall withstand an internal static air pressure of 1 psig for 30 minutes without leaking. Longitudinal and lateral baffles shall be incorporated to limit rapid shifting of agent; the beffles shall be no more than three feet on center. The baffles and other tank interior structures shall be designed to allow access to all parts of the tank for cleaning and maintenance and shall prevent vortices during pumping operations. Each tank shall be constructed to prevent complete penetration from corrosion or deterioration for 12 years when used as specified.

3.8.1.2 Agent tank mounting. The agents tanks shall be mounted in such a manner to minimize stress induced by frame flexing during off road operation.

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3.8.1.3 Agent tank vent and overflow system. Each agent tank shall incorporate vent system to relieve pressure on the tank during fill and discharge operations at maximum flow rates and an overflow system to relieve excess liquid in the event of tank overfill. The vent and overflow system shall prevent leakage of agent when the tank is filled to capacity and the vehicle is operating on the maximum side slopes and grades specified herein. Drainage from the vent and overflow system shall not flow over the body panels or other vehicle components and shall not be in front or behind any of the tires. Pipe or hose extensions shall be provided so that each point of discharge is approximately six inches above the bottom of the vehicle frame assembly. All tank drain and overflow system hoses shall be in accordance with Section 4-2 of NFPA 1961. Tank vent hoses shall be of the noncollaspible, wire reinforced type.

3.8.1.4 Agent tank drains. Each agent tank shall incorporate a drain, with a drain valve controlled by a crew member standing on the ground on the left side of the vehicle. The drain lines shall be two inch I.D. minimum. The point of discharge for the water tank drain shall be below the underbody panels. The foam tank drain outlet shall be located so that the contents of the tank, by use of a short section of hose, can be drained into 55 gallon drums.

3.8.1.5 Water tank manhole cover and fill opening. The water tank shall be equipped with an easily removable manhole cover with a minimum opening diameter of 20 inches. A top opening of not less than five inches internal diameter with a readily removable 0.25 inch strainer shall be provided. The fill opening may be incorporated as part of the manhole cover.

3.8.1.6 Water tank loading connections. The water tank shall incorporate a 4 1/2 inch male NSFHT hose connection and a 2 1/2 inch rocker lug female NSFHT hose connection on each side of the vehicle. The 4 1/2 inch loading connections shall permit filling from hydrants and other vehicles within two minutes at minimum pressures of 80 psi. The 2 1/2 inch loading connections shall permit filling from hydrants and other vehicles at an average rate of 375 GPM at a minimum pressure of 80 psi. The system design shall allow resupply of the water tank during pumping operations; a diffuser shall be provided to minimize turbulence. The connections shall be provided with chained-on aluminum alloy, long handled plugs and caps. The fill system shall be so designed and the lines so mounted to preclude accumlation of water at low ambient temperatures. The connections shall be no higher than 48 inches above the ground and readily accessible. The fill lines shall incorporate check valves and readily removable, rigidly constructed 0.25 inch mesh strainers.

3.8.1.7 Foam tank filler box. The foam tank shall incorporate a filler box mounted in the top of the tank readily accessible to at least two crew members on top of the vehicle. The filler box shall be at least six inches high and shall incorporate a hinged cover and latch, sealed so as to prevent spillage under any operating condition. The filler box shall be designed to allow two standard five gallon foam concentrate containers to be emptied simultaneously. The filler box neck shall extend sufficiently close to the bottom of the tank to reduce foaming to a minimum during fill operations. The filler box shall incorporate readily removable, rigidly constructed 10 mesh brass strainers and an easily visible marker to indicate when the tank is filled to capacity.

3.8.1.8 Foam tank loading connections. The foam tank shall incorporate a 1 1/2 inch NSFHT female hose connection on each side of the vehicle to permit filling by an external transfer transfer hose at flow rates up to 75 GPM. The connections sha

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be provided with chained-on aluminum alloy, long handled plugs. The connections shall be no higher than 48 inches above the ground and readily accessible. The fill lines shall incorporate rigidly constructed 0.25 inch mesh strainers.

3.8.1.9 Foam transfer pump. A self-priming positive displacement pump shall be mounted in a compartment on the vehicle. It shall be powered by the vehicle electrical, air, or hydraulic system. The pump shall be capable of transferring and drawing foam liquid concentrate at adjustable flow rates up to 75 GPM directly through the pump and loading connections (see 3.8.1.8). All materials and components which come into contact with the foam shall be compatible. The pump shall be removable as an assembly without disturbing other components.

3.8.2 Agent pump. The vehicle shall be equipped with a centrifugal pump capable of providing the performance specified herein. The suction inlet of the pump shall be lower than the bottom of the agent tanks to ensure gravity flow to the pump under all conditions specified herein. The pump shall be removable as an assembly without distrubing other components. The pump shall be provided with removable mechanical pump seals. Removal and replacement of the seals shall not exceed the maximum corrective maintenance downtime of four hours. The pump shall be constructed and smounted so that agent cannot enter any gearbox in the event of pump seal failure. The pump shaft shall be of 300 series stainless steel; housing and impellers shall be of brass or bronze. The pump shall be driven by a clutch designed to provide for smooth engagement and disengagement of the pump with the engine operating at any speed.

3.8.3 Agent system valves. Unless otherwise specified, agent system valves shall be of the quarter turn full on to full off type. All valves shall be mounted so as to be readily removable without disturbing piping and other components. All valves shall drain free in order to seal and to be operable without damage under the low ambient temperatures specified herein. All parts which could wear shall be replaceable or adjustable.

3.8.4 Agent system piping. Piping in contact with the agent shall be of brass, bronze or 300 series stainless steel. The agent system piping shall incorporate groove type couplings to the maximum practical extent. All piping sections between agent system components shall be readily removable without distrubing the components. Large pipe sections and pipe sections which could be subjected to excessive movement during operation shall be firmly mounted or braced. Sharp bends and changes in cross sections shall be avoided. The piping on the discharge side of the pump shall withstand a hydrostatic pressure of 150 percent of the maximum system operating pressure without leaking. The piping on the suction side of the pump shall be capable of maintaining a vacuum of at least 12 inches Hg, 10 minutes after being evacuated to 22 inches Hg vacuum. All pipe openings from heated compartments to the outside of the vehicle shall be provided with grommets or seals to limit air passage around the pipe. A check valve and shutoff valve shall be provided in each tank to pump line. A suitable and readily accessible central drain system shall be provided to permit draining the pump and all agent system piping within a four minute period. Piping for the 2 1/2 inch connections shall be 3 inch I.D. minimum and that for 1 1/2 inch connections shall be 1 3/4 inch I.D. minimum.

3.8.5 Pressure relief and overheat protection. The agent system shall incorporate a pressure relief valve to limit pressure on the discharge side of the pump to no more than 125 percent of the maximum system operating pressure. An automatic overheat protection system shall be provided to prevent agent temperature

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at the pump discharge from exceeding 170 degrees F with the pump at rated speed against a closed discharge for five minutes in any environmental condition specifie herein. Outlets from the pressure relief valve and the overheat protection system shall be directed to the ground. Surge suppressing devices shall be provided to prevent pressure in excess of the pressure relief valve setting during simultaneous shutdown of the turrets and the handline from maximum rated flow.

3.8.6 Foam proportioning system. The foam proportioning system shall be of the fixed or adjustable orifice type and shall automatically and uniformly proportion water and foam concentrate within the ratio of 2.8 percent to a maximum of 3.5 percent foam concentrate to water by volume. It shall also be capable of proportioning one percent foam concentrate (1.0 percent minimum, 1.8 percent maximum) and six percent foam concentrate (5.6 percent minimum, 7.0 percent maximum); additional orifice plates may be used for this purpose. The foam proportioning ratios for the 2 1/2 inch discharge connections shall be demonstrated at a flow rate of 250 GPM.

3.8.7 Roof turret. The roof turret shall be mounted near the front of the roof at the center of the vehicle. It shall have a non-air-aspirating, constant flow, variable stream nozzle with rates of 750 GPM (plus 75 GPM, minus 0 GPM) and 1,500 GPM (plus 150 GPM, minus 0 GPM) for foam or water. The discharge pattern shall be infinitely variable from straight stream to fully dispersed. The roof turret shall be provided with power assist, in-cab automatic oscillation, and manual override capability. The power assist mode shall be controlled by a joystick with a grooved pistol grip handle; power controls shall be positioned for use by the driver and the crew member seated to the right of the driver. For manual operation, the operating force shall not exceed 30 pounds for any discharge condition. The turret shall elevate at least 45 degrees above the horizontal and depress to allow agent discharge within 30 feet of the vehicle with a fully dispersed pattern. The turret shall rotate at least 120 degrees to either side of the vehicle longitudinal centerline. A turret elevation and azimuth indicator shall be provided and located inside the cab. The nozzle shall be designed to drain automatically when the discharge valve is in the OFF position.

3.8.8 Bumper turret. The vehicle shall be equipped with a constant flow non-air-aspirating, variable stream type bumper turret, mounted on the bumper in the front center of the vehicle. The bumper turret shall be capable of discharging at a flow rate of 250 GPM (plus 25 GPM, minus 0 GPM) of foam or water, with a pattern infinitely variable from straight stream to fully dispersed. The maximum reach of the straight stream shall be at least 150 feet from the tip of the nozzle. The bumper turret shall incorporate in-cab automatic oscillation and manual operation modes. The width of sweep shall be infinitely variable from 30 degrees to 180 degrees with the center of oscillation infinitely variable within the range of rotation. The elevation of the nozzle, and the rotational position in the manual operation mode, shall be controlled by a joystick with a grooved pistol grip handle located for use by the driver and the turret operator. The nozzle shall elevate at least 45 degrees above the horizontal and depress at least 20 degrees below the horizontal.

3.8.9 Hose reel. A hose reel with 150 feet of one inch hose in accordance with Section 8-4 of NFPA 1961 shall be located in a compartment on the left side of the vehicle (see 3.11.4). The handline nozzle shall be a constand flow, non-air-aspirating, variable stream type capable of discharging 60 GPM of either water or foam solution at 100 psi nozzle pressure. The straight stream shall have

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an effective reach of at least 60 feet. There shall be spinning teeth to ensure a uniform protective fog pattern. The nozzle shall be constructed of hard anodized aluminum and shall have a ball type shutoff with large horseshoe style handle and pistol grip. When using foam (in accordance with MIL-F-24385), the nozzle shall comply with NFPA 412 regarding expansion and drain downtime. Horizontal and vertical rollers shall be provided to reduce chafing of the hose. The reel shall incorporate an electric motor rewind capable of an average rewind speed of 75 FPM, a manual rewind for use in the event of electric rewind failure and an adjustable friction brake. An air purge system, with check valves to prevent water from entering the chassis air system, shall be provided. Controls for operation of the handline shall be mounted in or adjacent to the handline compartment so as to permit a crew member to operate the system and rewind the hose. All electrical components shall be protected against entry of water.

3.8.10 Agent discharge connections. Two 2 1/2 inch NSFHT male discharge connections shall be located on the left side of the vehicle and two on the right in the same relative position. Each connection shall incorporate a valve of the quarter turn ball type with twist to lock in any position. A liquid filled pressure gauge shall be recess mounted in the body adjacent to each valve so that the pressure to the hose can be determined and adjusted from outside the vehicle. The connections shall be no more than 48 inches above the ground. Each connection shall be provided with a chained-on rocker lug cap. All caps shall be constructed of aluminum alloy.

3.9 Halon 1211 system. The vehicle shall be equipped with a 500 pound capacity bromochlorodifluromethane (Halon 1211) fire fighting system. Halon 1211 shall be in accordance with MIL-B-38741. All metal components, except the agent tank, which come into contact with the agent shall be brass or anodized aluminum, unless otherwise approved by the procuring activity.

3.9.1 Halon 1211 agent tank. The Halon 1211 agent tank shall be designed and constructed in accordance with section VIII of the ASME Boiler and Pressure Vessel Code. Operating pressure shall be 225 psig, plus or minus 25 psig. The agent shall occupy 55 to 75 percent of the tank volume at 68 degrees F. The agent tank shall be mounted to permit removal and replacement; lifting provisions shall be provided.

3.9.1.1 Connectors and valves. The agent tank shall be equipped with a quick connect charging coupling, installed in the horizontal position. The tank shall also be equipped with an agent discharge valve, a tank blow-down valve, and a hose blow-down valve. These shall be ball valves of brass construction with teflon seats and seals. The nitrogen supply line connection shall be equipped with a check valve to prevent flow of agent into the pressure regulator.

3.9.1.2 Gauges. The agent tank shall be equipped with a pressure gauge and an agent level gauge. The gauges shall have waterproof dials; the pressure gauge shall be located adjacent to the Halon 1211 system hose reel, and the level gauge shall be located on the agent tank. With the appropriate compartment doors open, the gauges shall be readily visible by an operator standing on the ground.

3.9.2 Nitrogen pressure system. The nitrogen pressure system shall provide pressure for Halon 1211 discharge. The system shall include the nitrogen cylinder and its mounting, the nitrogen cylinder valve, a pressure regulator, a pressure gauge, a recharge connection and the necessary piping and hoses.

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3.9.2.1 Nitrogen cylinder. One DOT 3AA-2400, 220 cubic foot, compressed gas cylinder shall be provided, installed on rollers in a horizontal position for ease of replacement. Cylinder mounting shall permit cylinder replacement by two crew members standing on the ground in less than five minutes.

3.9.2.2 Nitrogen cylinder valve. A UL listed nitrogen cylinder valve shall be provided attached to the standard valve on the nitrogen cylinder. Provisions shall be made to actuate the nitrogen supply from inside the cab and from the handline compartment. The valve shall be a hand wheel type, with an integral lever operated quick opening device, an integral pressure relief valve and a pressure gauge. The valve shall permit instantaneous full gas flow from the cylinder by opening the hand wheel or pivoting the lever a maximum of 60 degrees. Each method of flow control shall be independant of the other method. All moving valve parts shall be hardened stainless steel, except for the camshaft. The valve shall be secured with a readily removable stainless steel safety clip; it shall be possible to install the shipping cap with the clip in place. The pressure relief valve shall limit pressure to no greater than 3,360 psi; a frangible disc shall not be used.

3.9.2.3 Pressure regulator. A pressure regulator shall be provided to reduce the supply pressure to the Halon 1211 system operating pressure. The regulator shall be designed for an inlet pressure of up to 3,360 psig and shall be permanently set to deliver an outlet pressure of 225 psig, plus or minus 25 psig, to the agent system within 10 seconds of opening the nitrogen cylinder valve. The regulator shall be equipped with a relief valve capable of handling the full gas flow of the regulator; the relief valve shall relieve at 325 psig, plus or minus 25 psig.

3.9.2.4 Nitrogen system pressure gauge. A pressure gauge with a waterproof dial shall be provided, located adjacent to the Halon 1211 agent tank pressure gauge in the Halon 1211 hose reel compartment. It shall be readily visible to a crew member standing on the ground.

3.9.3 Halon 1211 hose reel. A hose reel with 150 feet of one inch I.D. booster hose in accordance with UL 92 shall be located in a compartment on the right side of the vehicle. The reel shall incorporate an electric motor rewind capable of an average rewind speed of at least 75 FPM, a manual rewind for use in the event of electric rewind failure and an adjustable friction brake. Horizontal and vertical rollers shall be provided to reduce chafing of the hose. Controls for operation of the Halon 1211 system shall be mounted in the hose reel compartment. A nitrogen purge system shall be provided.

3.9.3.1 Halon 1211 nozzle. A nozzle with an on/off control valve shall be attached to the hose with a quick disconnect attached to a swivel connection that will allow full rotation. The nozzle shall be constructed of brass or anodized aluminum. The nozzle assembly shall be capable of discharging a visible conical pattern in fully open position and shall seal in the closed position with no noticeable leakage for one hour under maximum system operating pressure. A holder, located in the hose reel compartment, shall be provided for the nozzle.

3.9.4 Piping and hoses. All piping and hoses used in the Halon 1211 system, including the nitrogen pressure system, shall have proof pressure of not less than two times the maximum operating pressure.

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3.9.5 Leakage. The Halon agent system components shall not exceed the following maximum leakage rates:

AREA	MAXIMUM LEAKAGE RATE (Ounces/Year)
a. Welded or soldered seams (each seam)	0.6
b. Threaded connections (each connection)	5.0
c. Valves (each valve)	5.0
d. Gauges (each gauge)	5.0

3.10 Auxiliary equipment.

3.10.1 Optional equipment. Necessary compartments, holders, brackets and mounting pads shall be provided on all vehicles to accomodate the auxiliary equipment specified below in locations accessible to personnel standing on the ground. The holders, brackets, etc, shall be so designed and installed that the equipment is held firmly in place under all operating conditions specified herein and is readily removable for use. The following items shall be provided with the vehicle:

a. Two each 2 1/2 inch NSFHT straight double female hose connector in accordance with MIL-C-52404, Type XVIII, Class A.

b. One each 2 1/2 inch NSFHT female by 2 1/2 inch NSFHT male coupling in accordance with MIL-C-52404, Type XVII, Class C.

c. Two each 1 1/2 inch NSFHT straight double female hose connector in accordance with MIL-C-52404, Type XVII, Class A.

d. Two each 2 1/2 inch NSFHT female by 1 1/2 inch NSFHT male reducer in accordance with MIL-C-52404, Type XV, Class C.

e. One each dry chemical fire extinguisher in accordance with A-A-393, Class 1, Size 20, stored externally on the vehicle.

f. One each Halon 1211 fire extinguisher in accordance with A-A-1108, Type F, stored externally on the vehicle.

g. Two each spanner wrench in accordance with GGG-W-665, Type IV, stored in a compartment.

h. One each axe in accordance with GGG-A-926, Type II, stored externally on the vehicle.

i. One each Hooligan tool, Type TP30, stored on the vehicle.

j. One each rubber mallet in accordance with GGG-H-33, Type III, Class 3, stored in a compartment.

k. Two each portable automatic charging nicad type powered electric lantern complete with an automatic charging unit and an adjustable shoulder strap. The shoulder strap shall permit a crew member to carry the lantern with both hands free.

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3.10.2 Additional equipment. Additional compartments shall be provided for storage of the following fire fighting equipment in locations readily accessible to personnel standing on the ground. The additional equipment shall be furnished only with the first article test vehicles:

a. One each gasoline engine driven metal cutting saw kit with metal and carbide tipped blades. A cabinet with a minimum opening width of 38 inches and a minimum height of 19 inches shall be provided for the saw kit.

b. Two each Mine Safety Appliance Company Number 401 pressure demand Self-Contained Breathing Apparatus (SCBA) with storage case, one stored on each side of the vehicle.

c. Three hundred feet of three inch hose.

d. Three hundred feet of 1 3/4 inch fire hose (150 feet stored on each side of the vehicle).

e. One each 20 foot length of five inch soft suction hose with two each 4 1/2 inch female swivel connection.

3.10.3 Ladder. The vehicle shall be equipped with a 20 foot minimum, two section aluminum extension ladder in accordance with NFPA 1931. The ladder shall be securely top mounted or securely stored in a compartment on the right side of the vehicle. Easy access shall be provided for removal and replacement of the ladder.

3.10.4 Air hose reels. The vehicle shall be equipped with two automatic or electric rewind hose reel assemblies for the operation of pneumatic rescue tools. Each air hose shall consist of 150 feet minimum of 1/4 inch I.D. hose on a suitable reel assembly. The reel assembly shall have the capability to automatically rewind the hose when not in use. Air pressure to operate the tools shall be furnished by the vehicle air system, monitored by a regulator with a single or combined discharge of 6 CFM at 80 psig at maximum engine speed. The hose shall be provided with quick disconnect couplings. Each hose reel shall have a shut off valve. The hose reels shall be located in the compartments described in 3.8.9 and 3.9.3.

3.10.5 Back up alarm. An audible alarm shall be provided to caution personnel in the area when the vehicle is in reverse gear. The alarm shall be in accordance with SAE J994, Type C.

3.10.6 Air horn. The vehicle shall be equipped with a dual, electrically controlled, air actuated horn. The horn shall be mounted on the side of the cab or shall be roof mounted so as not to limit or restrict the operation of the vehicle.

3.11 Finish and protective coatings.

3.11.1 Cleaning, painting, plating, anodic films and chemical treatments. Cleaning, painting, plating, anodic films and chemical treatments shall be in accordance with MIL-STD-808. Unless otherwise approved by the procuring activity, primer shall be in accordance with MIL-P-23377.

3.11.2 Finish.

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3.11.2.1 Lime yellow. Unless otherwise specified (see 6.2), the outer surface of the vehicle shall be painted with DuPont Imron polyurethane paint, or equal, Lime Yellow, Color Number 13670 of FED-STD-595.

NOTE

'Equal' is defined as having performance identical to or superior to DuPont Imron when subjected to the following tests:

a. Test in accordance with MIL-C-46168:

(1) Paragraph 4.3.17 Flexibility.
(2) Paragraph 4.3.23 Accelerated weathering.

b. Test in accordance with MIL-C-83286:

(1) Paragraph 4.8.6 Low temperature resistance.
(2) Paragraph 4.8.10 Resistance to taping.
(3) Paragraph 4.8.12 Pencil hardness.

3.11.2.2 Forest green. When specified (see 6.2) the outer surface of the vehicle and the interior surface of compartments (except in the cab) shall be painted with polyurethane paint in accordance with MIL-C-83286, Forest Green, Color Number 24052 of FED-STD-595. All exterior trim (such as wheels, wheel covers, running boards, metal radiator grilles, bumpers, headlight trim rings, door handles and drip rails) shall be painted with the same paint as the exterior of the vehicle.

3.11.2.3 Desert sand. When specified (see 6.2) the outer surface of the vehicle and the interior surfaces of compartments (except in the cab) shall be painted with polyurethane paint in accordance with MIL-C-46168 or MIL-C-53039, Desert Sand, Color Number 30313 of FED-STD-595. All exterior trim items (such as wheels, wheel covers, running boards, metal radiator grilles, bumpers, headlight trim rings, door handles and drip rails) shall be painted with the same paint as the exterior of the vehicle.

3.11.3 Rustproofing. Unless otherwise specified (see 6.2) the vehicle shall be rustproofed in accordance with the contractor's standard commerical practice. The first article units shall not be rustproofed until after approval of the first article test report.

3.12 Markings. Unless otherwise specified, markings for vehicles painted Lime Yellow shall have reflectorized tape applied in accordance with L-S-300, Type I, Class 3, Reflectivity 1, white on a red background. Markings for vehicles painted Forest Green shall be applied with black paint, Color Number 37038 of FED-STD-595, and markings for vehicles painted Desert Sand shall be applied with white paint, Color Number 37875 of FED-STD-595.

3.12.1 Identification markings. Unless otherwise specified (see 6.2) the vehicle shall be marked for the appropriate service in accordance with MIL-STD-1223.

3.13 Identification plates and operating instructions. All identification plates, instruction plates, data plates, etc, shall conform to MIL-P-514 and be of a size and shape consistent with the information required thereon. Abbreviations shall be in accordance with MIL-STD-12.

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3.13.1 Identification plate. An identification plate, permanently marked in accordance with MIL-STD-130, shall be securely attached to the vehicle in a readily accessible location. The identification plate shall contain the following information.

NOMENCLATURE
MANUFACTURER'S MODEL NUMBER
MANUFACTURER'S SERIAL NUMBER
MANUFACTURER'S NAME
MANUFACTURER'S ADDRESS
MANUFACTURER'S SERVICE PHONE NUMBER
DATE OF MANUFACTURE
CONTRACT NUMBER
NATIONAL STOCK NUMBER

3.13.2 Control markings. All controls, valves, gauges and indicators used in the operation of the vehicle shall be identified by securely attached nameplates.

3.13.3 Operating instructions. Brief operating and precautionary instructions shall be permanently affixed near the appropriate controls. The instructions shall be clear, concise and adequate to enable operation of the vehicle without damage to the equipment or injury to personnel and shall refer to the components as identified by nameplates.

3.13.4 Transportation data plate. A transportation data plate shall be provided. The plate shall contain at least the following information:

a. Side and rear silhouette views of the vehicle.

b. Horizontal and vertical location of the center of gravity of the vehicle in transport condition.

c. Shipping weight.

d. Loading cubage.

e. Overall length, width and height.

f. Tiedown instructions for air transport in Types C-5 and C-17 Aircraft.

3.14 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. All parts shall be marked for identification in accordance with MIL-STD-130. (Note that MIL-STD-130 does not require marking of commonly known commercial parts which present no identification problem or parts in assemblies which are not normally subject to disassembly or repair.)

3.15 Dimensions. The vehicle, in fully operational condition, including all equipment and accessories, shall not exceed the following dimensions:

a. Length of 380 inches.
b. Width of 115 inches (except for mirrors).
c. Height of 150 inches.

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The vehicle shall have an angle of approach and an angle of departure of at least 30 degrees, an interaxle clearance angle of at least 12 degrees and an 18 inch minimum ground clearance to mid wheelbase.

3.16 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4.1 and 6.2.1).

3.17 Welders and Welding.

3.17.1 Welders. All welders employed in the fabrication of the vehicle, except for commercial components (see 6.3.2), shall be certified in accordance with the applicable requirements of MIL-STD-1595, the American Welding Society, or the American Society of Mechanical Engineers.

3.17.2 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease and other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the welded parts. Welds shall transmit stress without cracking or permanent distortion when the parts connected by the welds are subject to test, proof and service loadings.

3.18 Fasteners.

3.18.1 Bolt connections. Bolt holes shall be accurately punched or drilled and shall be deburred. Unless torque is specified, threaded fasteners shall be tight and shall not work loose during testing or service usage.

3.18.2 Riveted connections. Rivet holes shall be accurately punched or drilled and shall be deburred. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flatened, shall be of approved shape. Rivet heads shall be full, neatly made, concentric with the rivet holes and in full contact with the surface of the component.

3.19 Gear and lever assemblies. Gear and lever assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots or other irregularites. When required for accurate adjustment, gear assemblies shall be free of backlash.

3.20 Cleaning. The vehicle shall be thoroughly cleaned. Loose, spattered or excess solder, welding slag, stray bolts, nuts and washers, rust, metal particles, pipe compound and other foreign matter shall be removed during and after final assembly.

3.21 Workmanship. The vehicle, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs and sharp corners and edges; accuracy of dimensions, surface finish and radii of fillets; thoroughness of soldering, welding, brazing, painting, wiring and riveting; marking of parts and assemblies; alignment of parts; tightness of assembly fasteners; etc.

4. QUALITY ASSURANCE PROVISIONS.

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective materials, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4.1).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions.

4.3.1 Apparatus. Unless otherwise specified herein, all apparatus used in the testing and inspections specified herein shall be of laboratory precision type and shall be calibrated at proper intervals to ensure laboratory accuracy.

4.3.2 Data. During all testing specified herein, at least the following data, unless not applicable, shall be recorded at intervals not to exceed 30 minutes. Additional data and/or shorter intervals shall be provided as appropriate for any specific test.

a. Date.

b. Time started.

c. Time finished.

d. Ambient temperature.

e. Engine speed.

f. Engine oil pressure.

g. Engine coolant temperature.

h. Transmission oil temperature.

i. Vehicle odometer reading at the start and finish of each test.

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j. Engine hourmeter reading at the start and finish of each test.

k. Agent pressure at the pump outlet.

4.3.3 Preparation for test. The vehicle, equipped with all specified auxiliary equipment, shall be fully serviced and all equipment shall be prepared for immediate operation.

4.3.4 Test rejection criteria. During all tests specified herein, the vehicle shall be closely observed for the following conditions, which shall be cause for rejection:

a. Failure to conform to design or performance requirements specified herein.

b. Spillage or leakage of any liquid, including fuel, coolant, lubricant or agent, under any operating condition, except as allowed herein.

c. Structural failure of any component, including permanent deformation, or evidence of impending failure.

d. Evidence of excessive wear.

e. Interference between the vehicle components or between the vehicle, the ground, and all required obstacles, with the exception of normal contact by the tires.

f. Misalignment of components.

g. Evidence of undesirable roadability characteristics, including instability in handling during cornering, braking and while traversing rough terrain.

h. Conditions which present a safety hazard to personnel during operation, service or maintenance.

i. Overheating of the engine, transmission or any other vehicle component.

4.4 First article testing.

4.4.1 First article test samples. The first article test samples may be preproduction items or production items which meet the requirements of this specification (see 3.16, 4.2 and 6.2.1). The samples shall be identical to the items procured under the terms of the contract. Approval of the first article samples shall not constitute a waiver by the Government of its rights under the provisions of the contract. Two first article samples shall be produced and tested as follows:

a. Contractor Test Sample (see 4.4.2.1).

b. User Test Sample (see 4.4.2.2).

4.4.2 First article test.

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4.4.2.1 Contractor test sample. The contractor test sample shall be subjected to the tests described in 4.6.1 through 4.6.9.

4.4.2.2 User test sample. The user test sample shall be subjected to the tests in 4.6.1 through 4.6.2.5 prior to the test described in 4.6.10.

4.4.3. Test report. When specified in the contract or purchase order, and upon completion of the first article tests, a first article test report shall be prepared in accordance with MIL-STD-831 (see 6.2.2). An authorized Government representative shall countersign all data sheets and the report. The following should be incorporated as part of the test report:

a. All failures, maintenance performed and other events recorded shall be identified by accumulated operation time, miles, cycles or position in the test procedures, as appropriate. Test conditions at the time of failures or irregular operations identified shall be recorded.

b. Summary of the engineering reasoning and of any tests conducted to determine assignable causes for all failures and irregular operations identified.

c. Summary of the engineering reasoning behind any corrections made, to be made or proposed to be made, and the predicted effectivness of those corrections.

d. List of all cleaning and finishing procedures to be used in accordance with MIL-STD-808.

e. List of methods to be used to protect against electrolytic corrosion of dissimilar metals in intimate contact in accordance with MIL-STD-889.

f. Test report for each component covered by a document listed in section 2 of this specification.

4.5 Quality conformance testing. Quality conformance testing shall be classified as follows:

a. Individual Testing (see 4.5.1).

b. Random Sample Testing (see 4.5.2).

4.5.1 Individual testing. Each vehicle shall be subjected to the tests specified in 4.6.1 through 4.6.2.3 and 4.6.2.5.

4.5.2 Random sample testing. Any production vehicle may be selected by the procuring activity at any time to be subjected to the tests described in 4.6.1 through 4.6.4.3 and 4.6.7 through 4.6.7.1. Upon completion of the random sample testing, the contractor shall prepare a test report in accordance with MIL-STD-831. An authorized Government representative shall countersign all data sheets of the report (see 4.4.3).

4.5.2.1 Rejection and retest. When a vehicle fails to meet the requirements of this specification, vehicles to be delivered pursuant to the contract may not be accepted until the cause of the failure is determined, corrective action is taken by the contractor and the component that failed is successfully retested to assure

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compliance with this specification. Tests not involving the failed component may be continued pending the investigation of test failure.

4.6 Test methods.

4.6.1 Examination of product. The vehicle shall be inspected to determine compliance with the requirements specified herein. Particular attention shall be given to materials, workmanship, dimensions, surface finishes, protective coatings and their application, welding and riveting, and markings. For the first article and random sample vehicles, this examination shall be accomplished by the use of a checklist which lists each requirement not validated by a test and the results of the examination.

4.6.2 Preliminary operation.

4.6.2.1 Agent system pressure tests.

4.6.2.1.1 Agent tank hydrostatic pressure test. The agent tanks shall be hydrostatically pressure tested with water at a minimum of 1 psig for not less than 30 minutes. Minor weepage at the tank/tank top and tank vent connections is allowed; however, the development of water droplets shall be cause for rejection. Evidence of leakage shall be cause for rejection. This test may be performed prior to tank installation.

4.6.2.1.2 Agent system hydrostatic pressure and vacuum test. The agent system piping and all components on the pressure side of the pump shall be subjected to a hydrostatic pressure of 150 percent of the maximum system operating pressure for 15 minutes; evidence of leakage shall be cause for rejection. The agent system piping and all components on the suction side of the pump shall be subjected to a vacuum of 22 inches Hg. The vacuum shall not drop more than 10 inches Hg in 10 minutes.

4.6.2.1.3 Halon 1211 hydrostatic pressure test. With all pressure relief devices removed and openings plugged, the Halon 1211 agent tank and all components of the system that operate under pressure shall be hydrostatic pressure tested in accordance with Section VIII of the ASME Boiler and Pressure Code for Unfired Pressure Vessels. Each hose assembly shall be hydrostatically pressure tested at not less than two times the maximum operating pressure for at least 15 minutes. The initial and final pressures and evidence of impending failure shall be recorded (see 6.2.2). At the conclusion of the test, the fluid shall be completely drained and the components completely dried.

4.6.2.2 Halon 1211 leakage test. The agent tank shall be charged with a minimum of 50 pounds of Halon 1211 or refrigerant 12. The agent tank and all components which come into contact with Halon 1211 shall shall then be pressurized with nitrogen to 250 psig minimum and soaked at an ambient temperature of not less than 68 degrees F for four hours minimum. At the conclusion of the soak period, a Halogen leak detector shall be used to determine the leakage rate of each seam, connection, valve and gauge. Failure to comply with 3.10.5 shall be cause for rejection.

4.6.2.3 Road test. The vehicle, with the agent tanks approximately 75 percent full, shall be operated over hard surface highways for a distance of 100 miles at an average speed of at least 45 miles per hour. During the test, the following operations shall be performed:

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a. Normal or average stops, 10 times.

b. Quick stops from a speed of at least 30 miles per hour by instantaneous application of full service braking power, five times.

c. Sharp right turns and sharp left turns, five times each.

d. Operation through all forward and reverse gears, 10 times.

4.6.2.4 Cross country terrain operation. The vehicle, with the agent tanks approximately 75 percent full and tires with not more than 30 percent of original tread depth, shall be operated over cross country terrain for a distance of 30 miles at an average speed of at least 15 miles per hour. Tires may be new and shaved or worn from previous testing. The differential locking devices shall be engaged for at least one mile of operation.

4.6.2.5 Agent system operation. Each turret and the handline shall be fully tested separately and in combination for straight stream and fully dispersed patterns, for both water and foam discharge. The agent shall be observed to determine that there is no instability in proportioning, foam quality, pressure or flow rate under any operating condition. Each flow condition shall be maintained for at least 90 seconds.

4.6.3 Mobility test.

4.6.3.1 Maximum speed and acceleration. The time-speed-distance relationship of the fully loaded vehicle shall be determined while accelerating from a standing start to maximum speed on a dry level paved surface to demonstrate compliance with 3.2.1 (a) and (b). Not less than three runs, each in opposite directions, shall be conducted. Maximum speed and acceleration shall be the average for all runs.

4.6.3.2 Grade operation. The fully loaded vehicle shall be driven up and down 30 and 50 percent dry paved grade to demonstrate compliance with 3.2.1 (c). The maximum speed that can be maintained while climbing each grade shall be measured and recorded in the test report (see 4.4.3). Engine oil pressure shall be observed and shall not be less than the manufacturer's recommended minimum pressure at the engine operating speed. Engine starting ability shall be demonstrated in both directions on the grade.

4.6.3.3 Side slope operation. The fully loaded vehicle shall be driven in both directions on a side slope of 20 percent. Engine oil pressure shall be observed and shall not be less than the engine manufacturer's recommended pressure at the engine operating speed. Engine starting ability shall be demonstrated in both directions on the slope.

4.6.3.4 Ramp test. The fully loaded vehicle shall be driven forward from a hard, flat, horizontal surface and ascend a ramp with a minimum length of 1.2 times the vehicle wheelbase and a slope of 36 percent. It shall then proceed across the ramp crest onto an upper connecting hard horizontal surface until all tires are resting on the upper surface, down the ramp, and onto the lower horizontal surface. The test shall then be repeated in the opposite direction.

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4.6.3.5 Suspension articulation. The vehicle shall be tested to demonstrate compliance with 3.2.1 (l). The sides of the blocks may be sloped; however, the block length shall be such that when one tire on a bogie is at the top of the block, the other tire on that side is on the ground.

4.6.3.6 Wall to wall turning diameter measurement. The vehicle wall to wall turning diameter for turns in each direction shall be measured in accordance with SAE J695 to demonstrate compliance with 3.2.1 (m).

4.6.3.7 Brake test.

4.6.3.7.1 Service brake test. The vehicle shall be tested in accordance with SAE J786a to demonstrate compliance with 3.2.1 (j).

4.6.3.7.2 Parking brake test. The vehicle shall be tested in accordance with SAE J360 to demonstrate compliance with 3.2.1 (k).

4.6.3.8 Pump and roll operation. The vehicle shall be tested to demonstrate compliance with the pump and roll requirements of 3.2.1 (i). The vehicle shall maintain rated flow through the turrets, separately and in combination, for both water and foam discharge, for a minimum of 90 seconds for each discharge condition.

4.6.3.9 Vehicle cone index. The vehicle shall be tested to demonstrate compliance with 3.2.1 (n). During 10 separate passes in different tracks, the vehicle shall traverse terrain consisting of unflooded fine grained dry to moist soil having a cone index rating in the critical layer of no greater than 30. Testing shall be conducted using tires with not more than 30 percent of original tread depth. Tires may be new and shaved or worn from previous testing.

4.6.3.10 Ride quality. The vehicle shall be tested to demonstrate compliance with 3.2.1 (p). Testing shall be conducted using tires with no more than 30 percent of original tread depth. Tires may be new and shaved or worn from previous testing.

4.6.4 Capacity testing.

4.6.4.1 Agent tank fill.

4.6.4.1.1 Water tank top fill. The empty tank shall be filled with water from a 2 1/2 inch hose placed in the top fill opening at an average flow rate of at least 350 GPM. The quantity of water required to fill the tank shall be measured and recorded in the test report (see 6.2.2).

4.6.4.1.2 Water tank hydrant fill. The empty water tank shall be filled with water from a hose attached to each of the 2 1/2 inch fill connections at an average flow rate of at least 350 GPM and a pressure of at least 100 psig. The tank shall be filled from a hydrant using each of the 4 1/2 inch fill connections at a pressure of at least 100 psig to demonstrate compliance with the specified tank fill and overflow requirements.

4.6.4.1.3 Foam concentrate tank top fill. It shall be demonstrated that two, five gallon cans of foam concentrate can be emptied into the tank through the filler box by two operators in no more than 20 seconds. The quantity of foam concentrate required to fill the tank shall be measured and recorded in the test report (see 4.4.3).

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4.6.4.1.4 Foam concentrate side loading. The empty foam concentrate tank shall be filled with foam concentrate from a hose using each of the loading connections at the side of the vehicle at an average flow rate of at least 75 GPM.

4.6.4.2 Agent tank exhaustion. The capability to discharge the rated capacity of the water tank through the agent system when the vehicle is on level ground and 75 percent of tank capacity when the vehicle is headed in either direction on a 20 percent side slope and on a 30 percent grade shall be demonstrated.

4.6.4.3 Agent system discharge. The vehicle shall be tested to demonstrate compliance with the agent system performance requirements of 3.2.2.1, 3.2.2.1.1 and 3.8.6. Each turret and the handline shall be tested separately and in combination, for straight stream and fully dispersed patterns and for both water and foam discharge. Maximum pattern dimensions and ranges, based on uniform water application, shall be determined for the 100, 90, 80, and 70 percent outline areas. The foam from each nozzle shall be analyzed in accordance with NFPA 412. The agent system shall be observed to determine that there is no instability in proportioning, foam quality, pressure or flow rate under any operating condition. Each flow condition shall be maintained for at least 90 seconds.

4.6.4.4 Halon 1211 fire test. The Halon 1211 system shall extinguish three 600 square foot liquid fuel pan fires in accordance with UL 711. Neither Halon 1211 nor nitrogen shall be added to the system between fires.

4.6.5 Design verification.

4.6.5.1 Weight and weight distribution. The weight on each corner of the vehicle shall be measured with the unit standing on a flat level surface, both in the empty and fully loaded conditions. The fully loaded vehicle shall include ballast to simulate a 250 pound crew member in each seating position.

4.6.5.2 Dimensional verification. The dimensions specified herein shall be measured with the vehicle standing on a flat level surface. Overall height and body lean shall be measured in both the empty and fully loaded conditions, as defined in 4.6.5.1.

4.6.5.3 Electromagnetic interference measurement. The vehicle and all electrical components shall be tested in accordance with MIL-STD-462 to demonstrate compliance with 3.1.7.

4.6.5.4 Sound level testing. The vehicle shall be tested in accordance with SAE J336a and SAE J1077 to demonstrate compliance with 3.1.8.

4.6.5.5 Visibility. The limits of vision of the driver and the turret operator shall be determined by direct vision to demonstrate compliance with the visibility requirements of 3.4. Actual limits of vision shall be measured and recorded in the test report (see 4.4.3).

4.6.5.6 Dry pump operation. The agent system shall be completely drained and the agent pump shall be operated at rated pump speed for 10 minutes continuously. The agent system shall be filled, operated and checked for leaks. If there is evidence of pump damage, the pump shall be disassembled and inspected.

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4.6.5.7 Steering effort. The vehicle shall be tested to demonstrate compliance with the steering effort requirements of 3.3.2.

4.6.5.8 Relief valve test. Each pressure relief valve in the Halon 1211 system shall be subjected to a gradually increasing pressure of clean, dry, oil free air or nitrogen until its starts to discharge. The pressure shall than be reduced until the valve completely reseals. This procedure shall be repeated twice. The valve shall than be checked for leakage with its maximum working pressure applied. Failure to open or reseal within the specified limits or indication of leakage shall be cause for rejection.

4.6.5.9 Alternator output test. With the vehicle in neutral and the parking brakes engaged, the vehicle shall be tested to demonstrate compliance with 3.5.1. Operation, as described, shall be maintained for a minimum of one hour. There shall be no evidence of battery discharge.

4.6.6 Environmental tests.

4.6.6.1 Low temperature storage and operation. The vehicle shall be serviced with low temperature lubricants, and the agent tanks, pump and piping shall be drained. The vehicle shall be placed in an environmental chamber and subjected to the following tests:

a. The vehicle shall be subjected to an ambient temperature of 0 degrees F, plus or minus 2 degrees F, for a period of at least 24 hours. Upon completion of the soak period, it shall be demonstrated that the vehicle engine can be started in accordance with 3.3.1.1.4. The time required to start the engine shall be recorded in the test report (see 4.4.3).

b. The vehicle agent tanks shall be filled with agent no warmer than 60 degrees F. With the agent pump and all agent system piping filled, the winterization system shall be activated and the vehicle subjected to 24 hours at an ambient temperature of 40 degrees F, plus or minus 2 degrees F. At the conclusion of the soak period, it shall be demonstrated that the vehicle engine can be started in accordance with 3.3.1.1.4. The time required to start the engine shall be recorded in the test report (see 6.2.2). The heater and defroster shall be tested to demonstrate compliance with 3.4.6.1. The ability to discharge water through each outlet shall then be demonstrated. A hose shall be connected from a 2 1/2 inch discharge outlet to the 2 1/2 inch water tank loading connection on each side of the vehicle, and water shall be recirculated for two hours continuous pumping with a discharge outlet pressure of at least 100 psig. The winterization system shall be shut down and the vehicle subjected to an additional period of two hours at an ambient temperature of -40 degrees F, plus or minus 2 degrees F. Any evidence of freezing shall be cause for rejection.

The following temperatures shall be recorded at intervals not greater than 15 minutes during the low temperature storage and operation test (see 4.4.3):

a. Ambient.

b. Water in the water tank.

c. Foam in the foam concentrate tank.

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d. Air at the center and bottom of all compartments containing agent system components.

e. Surface temperature of all major agent system components.

f. Coolant inlet and outlet for all major components being heated (-40 degrees F storage and operation test only).

4.6.6.2 High temperature storage and operation. The vehicle shall be placed in an environmental chamber and subjected to an ambient temperature of 125 degrees F, plus or minus 2 degrees F, for a period of at least 24 hours. Upon completion of the soak period, the engine shall be started and agent shall be discharged at a rated flow through each outlet. A hose shall be connected to the 2 1/2 inch water tank loading connection on each side of the vehicle, and water shall be recirculated for two hours continuous pumping; pressure at the discharge outlet shall be at least 100 psig. The air conditioner shall be tested to demonstrate compliance with 3.4.5.2. Overheating of any component or evidence of impending failure shall be cause for rejection. The outlets shall be closed, and operation of the overheat protection system shall be demonstrated.

4.6.6.3 Rain. The complete vehicle, including all accessories, shall be tested in accordance with Test Method 506.2, Procedure III, of MIL-STD-810 to demonstrate compliance with 3.1.4 (c). Operational checkouts shall be in accordance with 4.6.6.5. The interior of the cab and all compartments shall be examined for water; any evidence of leakage shall be cause for rejection.

4.6.6.4 Other environmental test. The components and materials used in the construction of the vehicle shall be subjected to, or certified as having passed, or being capable of passing, the following test methods of MIL-STD-810 to demonstrate compliance with 3.1.4:

a. Humidity...Method 507.2, Procedure I.

b. Salt Fog...Method 509.2, Procedure I.

c. Sand and Dust...Method 510.2, Procedures I and II. For Procedure I, air velocity shall be 1750 ft/min and high temperature adjustment shall be 125 degrees F. For Procedure II, air velocity shall be 5700 ft/min; sand concentration of 0.0623 g/ft3, plus or minus 0.015 g/ft3; 90 minutes per face duration.

4.6.6.5 Operational checkout. All controls, levers, machanisms, and doors shall be actuated a sufficient number of times to ensure consistency of operation and to determine whether the test exposure produced system degradation. These actions are to result in the operation of the following systems, but will not be limited to:

a. Engine starting, operation and accessories.

b. Pumps; auxiliary pumps.

c. Climate control system, including defroster.

d. Cab doors and windows.

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e. Lights (i.e. headlamps, turn signals, gauges/controls, work lights, cab lights, etc).

f. Brakes, stationary (i.e. service, parking and emergency).

g. Windshield wipers.

h. Drivetrain (distance limited by chamber size).

i. Steering.

j. Equipment/compartment covers/doors.

k. Handrails/ladders/etc...removal/installation.

l. Miscellaneous equipment.

4.6.7 Reliability/durability testing. Compliance with the quantitative reliability requirements of 3.2.2.1 shall be demonstrated by the tests in 4.6.7.1 and 4.6.10. Apparent failures that are caused by some unintended factor external to the equipment (e.g. test equipment failure) shall not be counted as reliability failures if the cause is substantiated to the satisfaction of the procuring activity. Any test resulting in a failure shall be fully documented. Servicing, preventive maintenance or adjustments authorized by this specification or the procuring activity shall not be counted as failures.

4.6.7.1 Vehicle durability test. The vehicle shall be subjected to two cycles of operation at gross load, one cycle of operation with the agent tanks approximately 75 percent full, and one cycle of operation with the agent tanks empty over the following roads and terrain at the speed and for the distance specified:

PHASE	MINIMUM AVERAGE SPEED (Miles Per Hour)	DISTANCE (Miles)
Hard surface highways.	45	600
Gravel roads.	20	100
Cross country terrain.	15	200
Belgain Block.	15	100

At the conclusion of each loaded phase, the agent system shall be engaged and the vehicle operated in the pump and roll mode at maximum flow through the turrets until the water tank is empty. Engine fuel shall be Grade JP-4 turbine fuel.

4.6.8 Maintainability demonstration. Compliance with the maintenance downtime requirements shall be demonstrated by task performance. All scheduled maintenance tasks and all tasks listed in 3.1.3.2 shall be performed. In addition, corrective maintenance tasks shall be selected by the procuring activity from an approved list tasks provided by the contractor in the first article test procedure. Not less than 25 percent or more than 50 percent of the total number of tasks shall be demonstrated to verify an average maintenance downtime within the required limit. As part of this demonstration, the recommended frequencies of the scheduled maintenance tasks and the times recorded to accomplish the tasks shall be used to develop an expected value of scheduled maintenance downtime per measure of use, such as calendar time, hours of operation or miles traveled. The capability of

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performing maintenance tasks by personnel wearing arctic mittens shall be demonstrated.

4.6.9 Air transportability analysis. A detailed engineering analysis shall be conducted to demonstrate compliance with 3.1.9. This analysis shall include the tiedowns and all major components and their ability to withstand the accelerations specified in MIL-STD-8421. The evaluation shall also include a dimensional analysis for the vehicle while loaded aboard a Type C-17 Aircraft.

4.6.10 User test. The vehicle shall be delivered to the location specified (see 6.2) for 60 days of user operation. The test period shall begin after the vehicle has been inspected and prepared for use, and shall be for up to 480 hours of operation. The vehicle shall be available for any eight hour period specified by the Government each day, seven days a week. Any period of more than four hours that the vehicle is out of service for maintenance or repair shall not be considered part of the test period. The vehicle shall be operated, and scheduled maintenance shall be performed by Air Force personnel using the operation and maintenance procedures contained in the preliminary technical data to be supplied with the vehicle. Any parts or corrective actions required shall be provided by the contractor; all fuel, agents, oxygen and nitrogen shall be provided by the Air Force. Test operations will simulate normal Air Force usage and will not exceed any operating parameters defined by this specification. The sole criteria for acceptance/rejection of the user test shall be based on the performance requirements of this specification. During each day of operation, up to eight pan fires and simulated aircraft fires may be extinguished. The vehicle may be driven up to 200 miles per day over hard surfaces, gravel roads, and cross country terrain. Sudden stops, minimum radius turns, maximum speed and maximum acceleration may be included in the daily operation. The adequacy of vehicle lighting will be demonstrated during night operation. Any test described in this specification may be performed. All systems and subsystems of the vehicle may be tested at the beginning and the end of the test period. The contractor shall provide vehicle orientation for safe and efficient operation by Government personnel. The contractor shall record all data required for the first article test report (see 4.4.3).

4.7 Refurbishment. After successful completion of all required testing, each vehicle shall be restored to like new condition.

5. PACKAGING.

5.1 Preparation for delivery. Preparation for delivery shall as specified herein and in accordance with the terms of the contract.

6. Information for guidance only. (This section contains information of a general or explanatory nature which is helpful, but not mandatory.)

6.1 Intended use. This vehicle is intended for use as an aircraft crash fire rescue truck at fixed and forward airbases worldwide. It may be used singularly or in conjunction with other firefighting equipment. This vehicle is not suitable for use as a structural firefighting truck.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

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b. Heated rearview mirrors, if required (see 3.4.4).

c. Finish color required if not Lime Yellow, Color Number 13670 of FED-STD-595 (see 3.11.2.1).

d. Rustproofing, if other than the manufacturer's standard commerical practice, is required (see 3.11.3).

e. Identification markings required (see 3.12.1).

f. Location of user testing (see 4.6.10).

6.2.1 First article. When a first article inspection is required, the items should be first article test samples. The first article should consist of two units. The contracting officer should include specific instructions in the acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract (see 3.16 and 4.4.1).

6.2.2 Considerations of data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraph No.	Data requirement title	Applicable DID No.
4.4.3	First Article Test Report	UDI-T-23790

(Data Item Descriptions related to this specification, and identified in section 6 will be approved and listed in DOD 5010.12-L, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Definitions.

6.3.1 Belgian Block. A Belgian Block course is paved with unevenly laid granite or basalt block, forming an undulating surface. It duplicates a rough cobblestone road such as is found in may parts of the world. A cobblestone course may be used as an alternate.

6.3.2 Commercial. A manufacturer's standard (product), having been marketed, in production and sold in substantial quantities to the general public for a minimum of one year prior to the invitation for bid or request for proposal. An item is 'sold

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in substantial quantities" only when the quantities regulary sold are sufficient to constitute a real commerical market. Nominal quantities, such as models, samples, prototypes or experimental units, do not meet this requirement. The "general public" is a significant number of buyers other than the Government or affiliates of the offeror; the item involved must not be for Government end use. For the purpose of this specification, items acquired for "Government end use" include items acquired for foreign military sales.

6.3.3 Cross country terrain. Cross country terrain consists of open fields, broken ground and uneven terrain in general.

6.4 Subject term (key word) listing.

Firefighting, truck
Crash, rescue, firefighting, truck
Air transportable, crash, rescue, firefighting, truck
Diesel engine driven, firefighting, truck
Aircraft, crash, rescue, firefighting, truck
Fixed and forward airbase, crash, rescue, firefighting, turck
Rough terrain, crash, rescue, firefighting, truck
Halon 1211, crash, rescue, firefighting, truck
Foam/water, crash, rescue, firefighting, truck
Agent, Halon/water, crash, rescue, firefighting, truck

6.4 Changes from previous issue. Asterisks (or verticle lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Air Force-99

Preparing activity:
Air Force-84

Agent:
Air Force-99
(Project 4210-F223)

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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